

Revisiting the Internationalisation-Performance Relationship in an Emerging Market: The Roles of Technological Capability and Digital Adoption

This study analyzes the relationship between internationalization and firm performance in Vietnam using World Bank Enterprise Survey data from 2009, 2015, and 2023. It finds a robustly nonlinear (inverted-U) relationship between export intensity and labor productivity, which is moderated by technological capability and digital adoption. The paper's main contribution is to show that the performance impact of these capabilities, particularly digital adoption, is not static but stage-contingent, evolving across different phases of the country's economic and digital transition.

Overall Feedback

Digital adoption construct

An apparently critical issue is that the paper's digital-adoption construct is more specific than several of the claims built around it. The primary DAI_z is a within-wave standardized website-presence dummy, while the abstract, §§2.3–2.4, §§5.1–5.5, and H3/H4 often speak in broader terms: digital adoption, transaction support, digital capability, and sometimes a digital-transformation channel. That broader language gives the construct more theoretical reach than the harmonized WBES measure can securely carry.

There is also a substantive consistency issue between §2.3 and §3.2. Section 2.3 says the cross-wave DAI_z is anchored in "website presence, foreign-licensed technology," whereas §3.2 assigns $e6$ exclusively to TCI_z and defines DAI_z as $c22b$ alone. Because construct separation is one of the paper's stated contributions, this needs to be made completely unambiguous.

The easiest way to strengthen the paper may be to re-theorize DAI_z explicitly as website-based digital presence or foundational website adoption, and then remove language about transaction support, process integration, and broader digital transformation from the main claims. The 2023 $e6$ payment measures can still be valuable, but they would land better as a clearly exploratory 2023-only extension through DAI_{rich} , not as validation of the harmonized cross-wave construct.

A related interpretive point is temporal. From 2009 to 2023, having a website plausibly changes meaning: in 2009 it may signal a relatively distinctive external-market interface; by 2023 it may be a more routine marker of basic digital presence. The stage-contingency interpretation currently tends to read changes in DAI_z coefficients as changes in the value of digital adoption itself. Readers may also see part of that pattern as proxy obsolescence. Naming this possibility would make the lifecycle discussion more defensible.

Technological capability construct

The paper's technological-capability argument is theoretically rich, but the primary measure is relatively thin. Section 2.2 presents technological capability as absorptive capacity, learning,

problem-solving, process upgrading, and broader organizational competence, drawing on Cohen and Levinthal (1990), Lall (1992), Helfat and Peteraf (2003), and Teece (2007). Section 3.2, however, operationalizes the primary TCI_z as the mean of internationally recognized quality certification and foreign-licensed technology.

Those two items are relevant, but they may capture standards compliance, foreign-market linkage, or exposure to foreign technology as much as firm-internal capability stocks. They may also be closely connected to exporting itself, which matters because the paper asks TCI_z to explain productivity differences around direct export intensity.

The concern becomes sharper in §4.5 Panel A. When product innovation and R&D are added to form TCI_{full} , the 2015 direct coefficient becomes insignificant and the moderation tests are non-significant in both 2015 and 2023. The current interpretation that TCI_{thin} is “sharper” may not be enough for reviewers, because the attenuation raises a deeper question: is the main result about technological capability broadly, or about foreign technology and quality-standards exposure more specifically?

One possible revision would be to narrow the label and theory around the primary construct, presenting TCI_z as foreign-technology/standards capability rather than as a general technological-capability stock. Another would be to keep the broader theory but treat the TCI_{full} results as a genuine boundary condition on the capability interpretation rather than mainly as support for the thinner measure.

Cross-wave interpretation

The lifecycle reading is one of the most interesting parts of the paper, but it currently runs ahead of the formal cross-wave evidence. The introduction says the three waves make a lifecycle reading “testable,” and §§4–5 interpret 2009, 2015, and 2023 as distinct stages of Vietnam’s institutional and digital transition. That framing is plausible and attractive, but the design remains repeated cross-sectional: the firm composition changes across waves, sector controls are broad one-digit categories, and no firm-level maturation is observed.

The paper’s own §4.5 Panel F also complicates the stronger lifecycle claim. The formal pairwise cross-wave tests show that all $FSTS_c$, $FSTS_c^2$, $FSTS_c \times DAI_z$, and $FSTS_c^2 \times DAI_z$ differences are non-significant. The clearest formal cross-wave evidence concerns the direct DAI_z coefficient: the 2009-to-2015 drop and 2015-to-2023 recovery are statistically distinguishable. That is useful evidence of wave-specific variation in website-presence associations, but it is thinner evidence for structural stage contingency in the internationalization-performance curve or in digital moderation.

If the paper wants to retain a strong lifecycle claim, it would help to add common-specification evidence that directly tests cross-wave differences: pooled wave-by-focal-variable interactions, composition-adjusted comparisons, and common-sector or more tightly harmonized sector analyses. If those tests remain limited, the claims could be recalibrated to wave-specific associations that are consistent with stage contingency rather than demonstrating it. That shift would preserve the contribution while reducing the risk that reviewers see the lifecycle language as stronger than the design permits.

Inverted-U validation

The inverted-U result is central to H1 and to the paper's broader framing, so it would benefit from more validation against the measurement structure of internationalization. Section 3.2 measures internationalization only as direct-export intensity, $FSTS = d3c/100$, with all non-direct exporters coded at zero. Table 1 shows that exporter shares are 28.4%, 20.8%, and 18.8% across the three waves. That means the quadratic in $FSTS$ is being estimated on a bounded, zero-inflated variable with a large participation margin at zero.

The Lind-Mehlum tests and turning-point confidence intervals are useful, and the 39–46% turning points are reassuring in one respect because they sit inside the observed range. But those tests do not by themselves show whether there is enough observed density around the turning points, or whether the fitted curve is driven substantially by the zero/nonzero contrast between non-exporters and exporters.

Before the inverted-U carries the theoretical weight assigned to it, readers may want to see that the pattern survives specifications that separate export participation from export intensity. Exporter-only estimates, two-part or hurdle-style specifications, spline or binned export-intensity models, and distributional evidence around the 39–46% turning points would all help. Even a compact diagnostic figure or table showing where observations sit relative to the estimated turning points would make the nonlinear claim easier to trust.

Moderation evidence

The digital-moderation contribution should be framed with more caution unless the statistical signal is strengthened. H4 and the originality claim depend heavily on DAI_z moderation, but the evidence currently reported is marginal: in 2023 the M8 joint test is $p = .062$, the pooled M8 joint test is $p = .083$, and the richer 2023 DAI_{rich} checks in §§4.1–4.5 are weaker. In §4.2, the pooled direct DAI_z coefficient also becomes null once interactions are introduced, which complicates the claim that DAI_z is positively associated with performance on average.

A further presentation issue is that the interaction model includes $FSTS_c$, $FSTS_c^2$, DAI_z , $FSTS_c \times DAI_z$, and $FSTS_c^2 \times DAI_z$. In that setting, coefficient signs and joint p-values alone do not show how the internationalization-performance curve changes across the observed export-intensity range. Reviewers will likely want to see the implied curves, not just the interaction coefficients.

Predicted curves or marginal-effect plots for low and high DAI_z and TCI_z would clarify the substantive pattern. Tests of slope differences and turning-point differences would also help distinguish whether the moderation changes the height, steepness, or location of the curve. Given the number of related tests across waves, constructs, and robustness panels, clearer multiple-testing caution would also improve the credibility of the interpretation.

If the signal remains at the current level, H4 may be better positioned as an exploratory or emerging pattern concentrated in 2023 rather than as a central confirmed contribution. That would still leave the paper with a strong core: a repeated-cross-section analysis showing a nonlinear direct-export-productivity association, a positive foreign-technology/standards capability association, and suggestive evidence that website-based digital presence becomes more conditional in the latest wave.

#1 Discrepancy in the 2015 and pooled sample sizes

 *Figure 2b. Predicted $\ln$ (labour productivity) across direct-export intensity, Vietnam 2015 ($N=956$). The fitted curve comes from the M2 inverted-U OLS specification with HC1 robust standard errors*

The reported sample sizes appear to be internally inconsistent. §§3.1–3.2 and the descriptive table report $N = 958$ for 2015 and $N = 2,960$ pooled, whereas the abstract, Figures 2b/2d, Table LM, and Table 3 use $N = 956$ and $N = 2,958$. The paper should either state one analytic sample definition consistently or explain why different sections use different N values.

#2 Misinterpretation of IMR vs. generalized residual

 *The inverse Mills ratio added to the outcome equation estimated on exporters ($FSTS > 0$) is statistically insignificant in all four panels: $\lambda = -0.111$ ($p = .846$) in 2009, $\lambda = -0.833$ ($p = .259$) in 2015, $\lambda = 0.665$ ($p = .399$) in 2023, and $\lambda = -0.462$ ($p = .347$) pooled, indicating no significant selection bias along the IMR-detected channel. A complementary control-function specification using the generalised residual yields a significant residual in 2009 ($\beta = 0.510$, $p < .001$) and pooled ($\beta = 0.304$, $p < .001$), a marginal residual in 2023 ($\beta = 0.259$, $p = .092$) and a null residual in 2015 ($\beta = -0.112$, $p = .357$). The control-function signal indicates a degree of selection structure that the standard IMR does not pick up*

The comparison between the IMR and the generalised-residual control function needs to be interpreted more carefully. In a probit selection model, the generalised residual coincides with the IMR for selected observations, so the reported difference in significance is more plausibly due to the correction being used in a different sample or specification than to the residual detecting a distinct selection channel. As written, Panel E overstates what this contrast establishes and should be framed as evidence about selection into any exporting rather than a full correction for endogeneity in $FSTS_c$ and $FSTS_c^2$.

#3 Contradictory p-value in Table 3 Panel C

*| B. DAI_{rich} binary (2023) | 1,013 | $FSTS_c \times DAI_{rich_bin_z} = -0.893$ ($p = .169$) n.s. | M8 joint $p = .243$ n.s. | | C. Common-N comparison (2023) | 1,013 | DAI_z moderation reestimated on DAI_{rich} sample | M8 joint $p = .013$ * |*

Table 3 Panel C appears inconsistent with the main 2023 results. The paper elsewhere reports the 2023 DAI_z M8 joint test as $p = .062$, but Panel C gives $p = .013$ even though it is described as re-estimating the same DAI_z moderation model on the same $N = 1,013$ sample. Unless Panel C is actually a different specification, the reported joint p-value looks incorrect or mislabeled.

#4 Contradiction regarding stage-contingent moderation

All $FSTS_c$, $FSTS_c^2$, $FSTS_c \times DAI_z$ and $FSTS_c^2 \times DAI_z$ pairwise differences are non-significant (all $|z| < 1.4$, all $p > .17$), so the inverted-U shape and the DAI moderation patterns are best read as a common curvature with wavespecific magnitudes rather than as cross-period structural shifts in the curvature parameters.

The evidence for stage contingency is stronger for the direct DAI_z association than for the moderation terms. Because Panel F finds no statistically significant cross-wave differences in $FSTS_c \times DAI_z$ or $FSTS_c^2 \times DAI_z$, it weakens the claim of a wave-specific shift in DAI moderation (H4). The claim that the moderation channel itself changes structurally or emerges only in 2023 should be stated more cautiously, even if the broader lifecycle pattern in digital adoption remains plausible. Specifically, the 2023 result is better described as the only wave in which moderation is detectable within-sample, not as clear evidence of a cross-wave structural change in the moderation coefficients.

#5 Contradictory coefficient for 2015 quadratic term

The 2015 wave captures a transitional phase in which the I-P curvature is unusually sharp ($FSTS_c^2 = -2.082$, $p = .005$) but the digital channel compresses entirely: DAI_z loses direct salience and shows no joint moderation

The 2015 quadratic term is reported inconsistently within §4.1: the earlier M2 discussion and Table 1 give -2.115 ($p = .004$), while this institutional-reading paragraph gives -2.082 ($p = .005$). If this later number comes from a different specification, that shift should be identified explicitly; otherwise one of the reported values is incorrect.

#6 Contradiction regarding exporter cohort expansion

DAI_z loses direct salience and shows no joint moderation, suggesting that the firms added to the exporter cohort during this period derived their productivity gains from scale and from technological capability rather than from foundational digital adoption.

There seems to be an issue with the phrase "firms added to the exporter cohort": with repeated cross-sections, the paper cannot identify actual entry into the exporter cohort, only differences in exporter composition across waves.